Redmi 6 & Redmi 6A Grade III Maintenance Guide V01

Technical support internal document control: TSIMHNC3C & C3D Redmi 6 & Redmi 6A three-level maintenance guidance V01

Scope of application:

Analysis center, mainboard and whole machine maintenance factory

Change history:

First edition 2018-09-10

1. Introduction of basic information

1.1 Product Overview

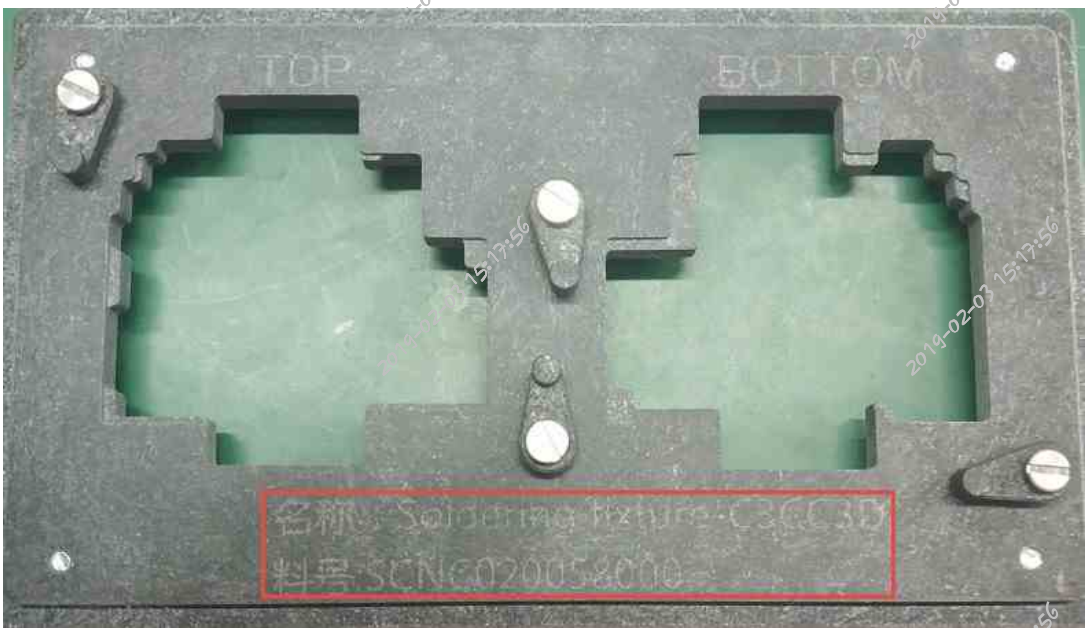
Product Overview:

红米6（C3D）
处理器与内存
3+32G & 3+64G
Helio P22八核处理器，12nm制程工艺，最高主频2.0GHz
摄像头
后置1200万+500万 AI 双摄
前置500万
指纹
后置指纹识别
续航
3000mAh（typ） / 2900mAh（min）
内置电池，免更换
支持5V1A充电
全网通5.0 双卡双待双4G
全网通5.0，支持 Nano-SIM卡/Micro-SD 扩展存储卡，支持双卡不限运营商，均可4G驻网；
双Nano-SIM卡槽，任意卡槽均可设置为主卡，支持移动/联通/电信 4G/3G/2G，支持VoLTE高清语音
网络频段
GSM（频段 B2/3/5/8）
CDMA（频段 BC0）
CDMA EVDO（频段 BC0）
WCDMA（频段B1/2/5/8）
TD-SCDMA（频段 B34/39）
TDD-LTE（频段 B34/B38/39/40/41（100MHz））
FDD-LTE（频段 B1/3/5/7/8）

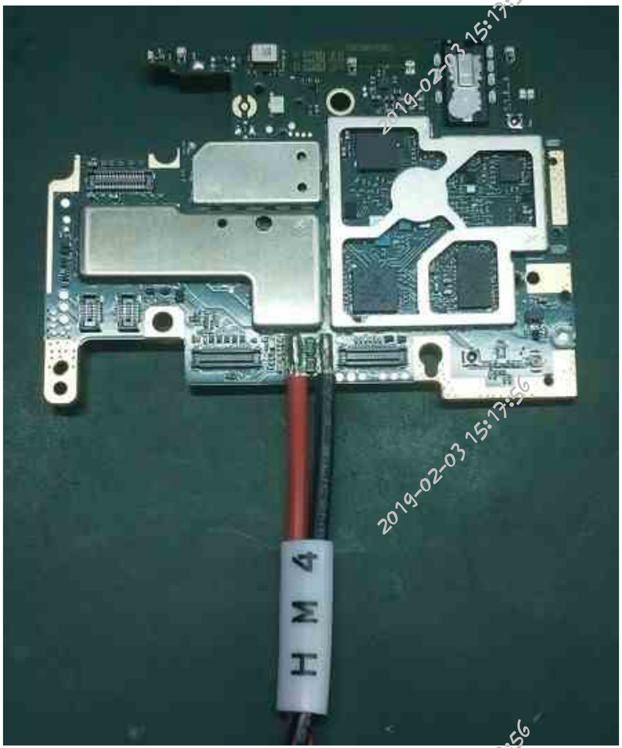
红米6A（C3C）
处理器与内存
2+16G &3+32G
Helio A22 处理器，12nm 制程工艺，4核A53架构，主频高达2.0GHz
摄像头
后置1300万像素单摄
前置500万像素
指纹
无
续航
3000mAh（typ） / 2900mAh（min）
内置电池，免更换
支持5V / 1A充电
全网通5.0 支持双卡双待双4G
全网通5.0，支持 Nano-SIM卡/Micro-SD 扩展存储卡，支持双卡不限运营商，均可4G驻网。
双Nano-SIM卡槽，任意卡槽均可设置为主卡，支持移动 / 联通 / 电信 4G/3G/2G，支持VoLTE高清语音
网络频段
GSM（频段 B2/3/5/8）
CDMA1X / EVDO（频段 BC0）
WCDMA（频段B1/2/5/8）
TD-SCDMA（频段 B34/39）
TDD-LTE（频段 B34/B38/39/40/41（100MHz））
FDD-LTE（频段 B1/3/5/7/8）

1.2

Material code: SCNC020058000



The false wires of Redmi 6 & Redmi 6A are the same as those of Redmi 4. Material code: SCNC020022700



1.4 Pasting position and specification of maintenance label

Sticking position of maintenance label: Stick on the shielding shell of auxiliary power supply.

Pasting standard: paste transversely aligned along the left edge and lower edge of the secondary power shield.

1.5 Matters needing attention in motherboard maintenance

Note:

1. After replacing EMMC, it can be turned on normally in factory mode. If TEEKEY is not uploaded, it will be turned on and restarted in user mode.
2. Take anti-welding measures when welding components near plastic parts such as key interface, display interface and earphone interface, which are easy to weld.

1.7 Brush mode

Brush platform: SP Flash Tool
Brushing tool: Redmi 6MTK separate
brushing tool Tool version: V5.1812. 0
Redmi 6A factory package:
SW_S98506AA1_V039_M13_XM_C3C_USRD_ATO_2
Redmi 6 factory package:
SW_S98507AA1_V039_M13_XM_C3D_USRD_ATO_2

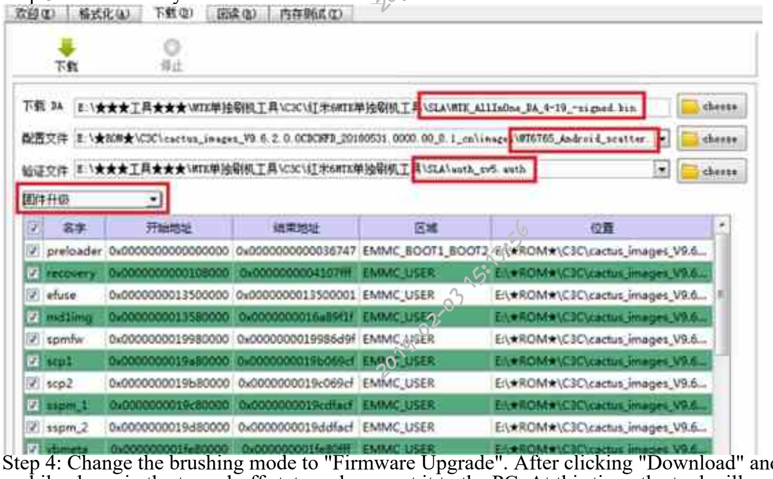
Brushing method

Step 1: Open the SP Flash Tool brushing tool, click "Download DA" to select the DA file corresponding to the model: Redmi 6, Redmi 6A, select "MTK_AllInOne_DA_4-19_signed. bin" under the tool folder;

Step 2: Click "Configuration File" to select "MTXXXXX_Android_scatter.txt" file in the machine brushing package. After selection, the tool will pop up the Processing interface and wait to enter

The next operation can be carried out after the degree bar is finished;

Step 3: Click "Verify File" and select "auth_sv5. auth" under the Tools folder



Step 4: Change the brushing mode to "Firmware Upgrade". After clicking "Download" and getting ready, press and hold the volume button on the mobile phone in the turned-off state and connect it to the PC. At this time, the tool will pop up the Xiaomi account login window, and log in to the Xiaomi account with deep brushing permission to start normal brushing (after the first login is completed, you can choose automatic login later)

When waiting for the brush to be completed, the prompt window of download completion will pop up in the tool interface as follows:

Remarks: For motherboards that have been replaced with CPU and EMMC or have been replaced with EMMC alone, the brushing mode should be changed to "all formatted and downloaded".

1.8 RF calibration test correlation

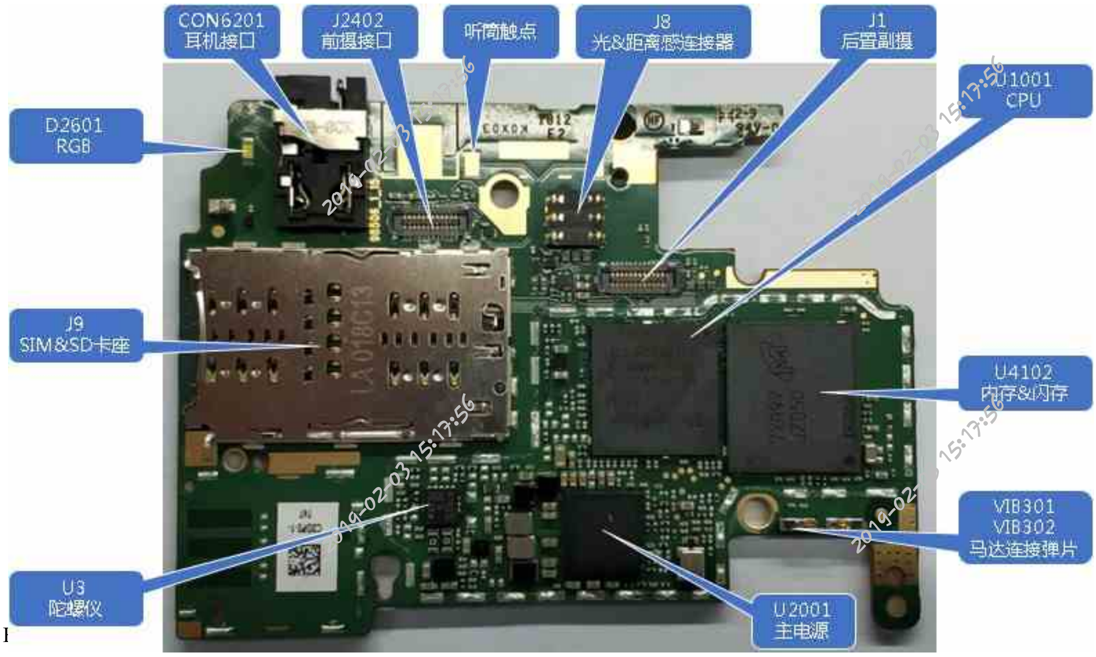
1.Open the calibration software, click START first, connect the main receiving antenna and auxiliary receiving radio frequency lines, and then use USB cable to connect the computer and the mobile phone, and the radio frequency calibration of the mobile phone begins.

2. After successful calibration, you can see that "PASS" is displayed in all frequency bands of RF in the version information in factory mode.
3. After successful calibration, the card can be directly inserted to test the signal.

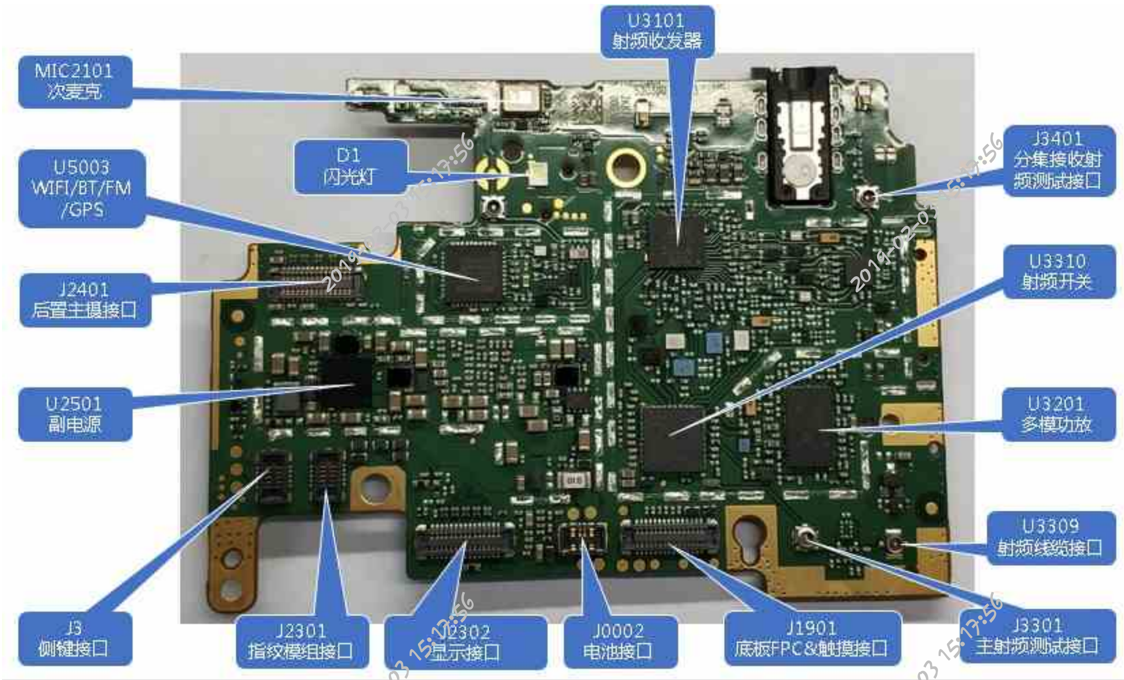
2. Brief introduction of motherboard module

2.1 Redmi 6 & Redmi 6A Main Board Component Distribution Diagram and Difference

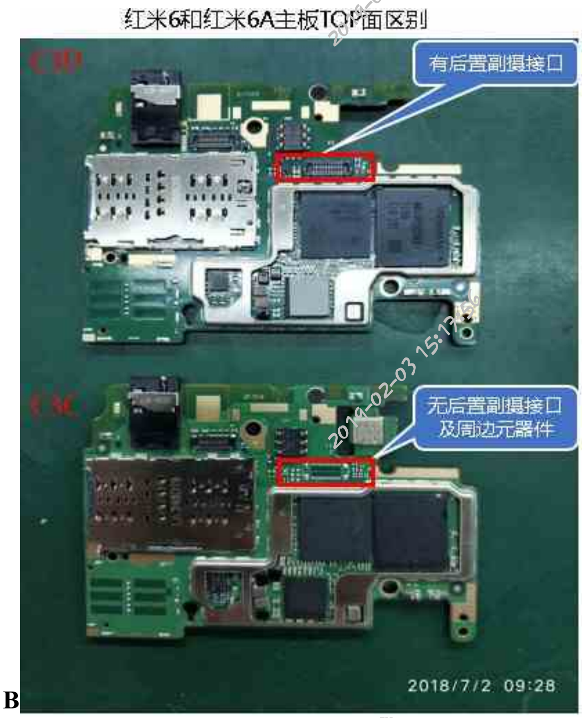
- TOP plane

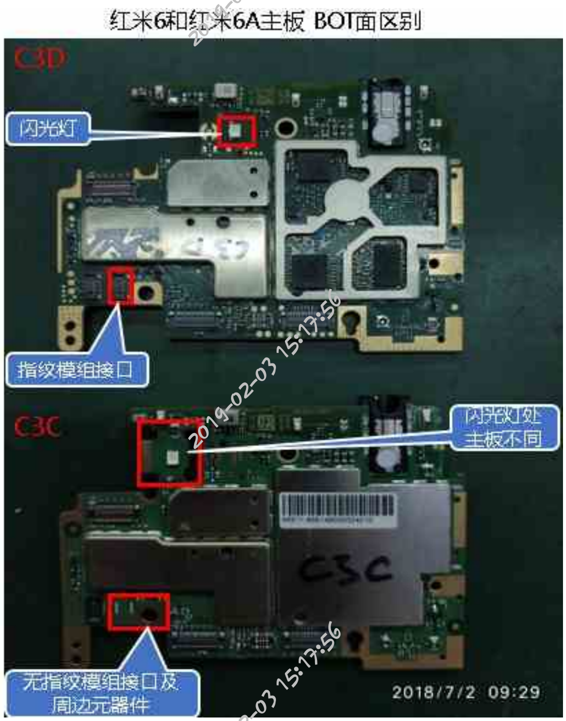


- I



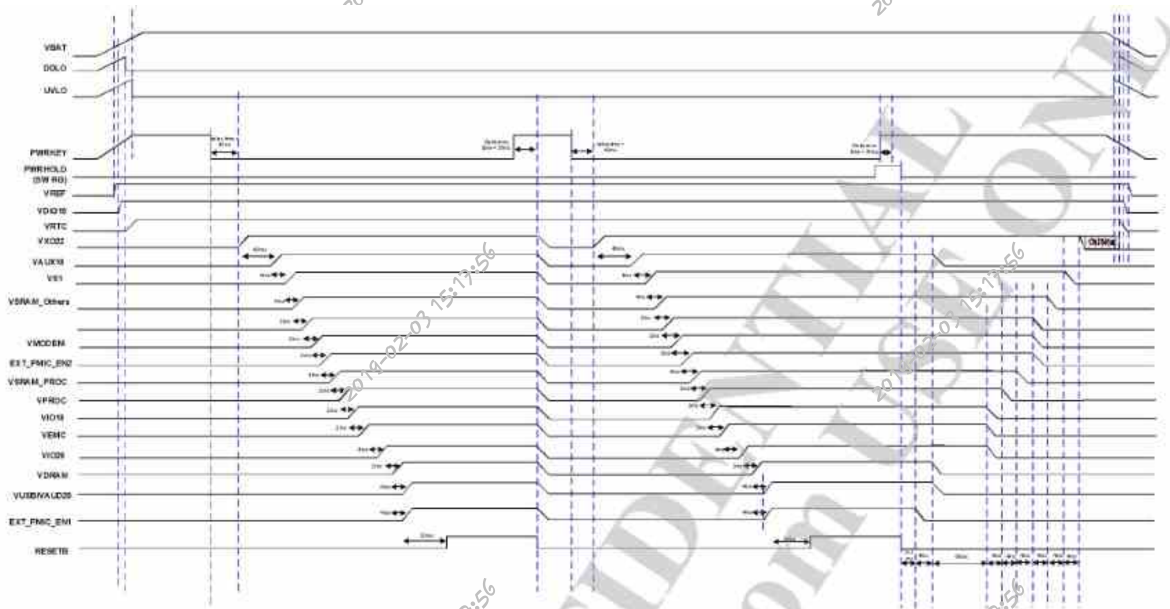
- Motherboard difference





2.2 B

Boot timing diagram:



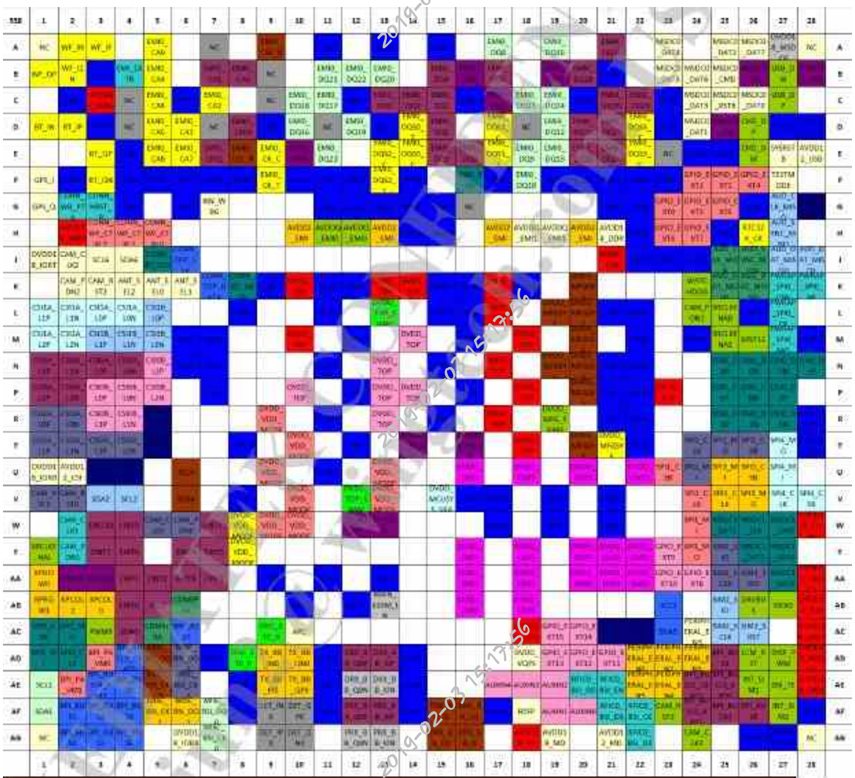
Note: Those timings are typical values. Timing variation is +-20%(Charger-plug-in/out related delay timing variation is +-40%).

B

开机时序测量表		
Symbol	**测量值**	**测量点**
PWRKEY	3.9V	R2703
VREF	1.1V	C2153
VRTC	32.768KHz	TP47
VXO22	2.2V	C2117
VAUX18	1.8V	C2114
VS1	1.8V	PL2005
VSRAM_Others	0.58V	C2156
VMODEM	0.53V	PL2003
VSRAM_PROC	1V（开机时有）	C2157
VIO18	1.8V	C2139
VIO28	2.8V	C2134
VDRAM	1.17V	C2161
VUSB/VAUD28	2.8V	C1165

2.3 Redmi 6-point bitmap

U1001 chip bitmap:



221Ball FBGA														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	DNU	VSF	VSS_m	VCCQ_m	DAT6_m	CMD_m	RCLK_m	VSS_m	DAT0_m	DAT5_m	VDDI_m	VSS_m	VSF	DNU
B	VSF	VSS_m	VCC_m	DAT7_m	DAT3_m	VCCQ_m	VSS_m	CLK_m	VCCQ_m	DAT1_m	VSS_m	VCC_m	VCC_m	VSF
C		RST_m	VSS_m	VCC_m	VSS_m	DAT2_m	VCCQ_m	VSS_m	DAT4_m	VSS_m	VCCQ_m	VSS_m	VSS_m	
D		VSF	VSF	VSF	VSF	VSF	VSS_m	VCC_m						
E														
F		VSS_v	VDD1_v	VDD1_v	VDD2_v			VDD2_v	VDD1_v	DQ29_v	DQ30_v	DQ31_v	VSSQ_v	
G		ZQ0_v	ZQ1_v	VSS_v	VDD1_v			VSS_v	VDDQ_v	DQ26_v	VSSQ_v	DQ27_v	DQ28_v	
H		CA9_v	VSS_v	VSSCA_v	VSS_v			VDDQ_v	DQS3_t_v	VSSQ_v	DQ24_v	VDDQ_v	DQ25_v	
J		CA8_v	CA7_v	VSSCA_v	VDD2_v			VSSQ_v	DQS3_c_v	DM3_v	VDDQ_v	DQ15_v	VSSQ_v	
K		VDDCA_v	CA6_v	VSSCA_v	VDD2_v			VSSQ_v	VSSQ_v	VDDQ_v	DQ13_v	VDDQ_v	DQ14_v	
L		VDD2_v	CA5_v	VSS_v	VDD2_v			VDD2_v	VDDQ_v	VSSQ_v	DQ12_v	VSSQ_v	DQ11_v	
M		VREFCA_v	VSS_v	VDD_v	VDD2_v			VSSQ_v	DQS1_t_v	VDDQ_v	DQ10_v	VDDQ_v	DQ9_v	
N		VDDCA_v	CK_c_v	VSS_v	VDD2_v			VSS_v	DQS1_c_v	DM1_v	VDDQ_v	DQ8_v	VSSQ_v	
P		VSSCA_v	CK_t_v	VSS_v	VDD2_v			VDD2_v	VSSQ_v	ODT	VDD2_v	VSS_v	VREFDQ_v	
R		CKE1_v	VSS_v	VSS_v	VDD2_v			VSS_v	DQS0_c_v	DM0_v	VDDQ_v	DQ7_v	VSSQ_v	
T		CKE0_v	CS1_c_v	VSS_v	VDD2_v			VSSQ_v	DQS0_t_v	VDDQ_v	DQ5_v	VDDQ_v	DQ6_v	
U		VDDCA_v	CS0_c_v	VSSCA_v	VDD2_v			VDDQ_v	VDDQ_v	VSSQ_v	DQ3_v	VSSQ_v	DQ4_v	
V		VDDCA_v	CA4_v	VSSCA_v	VDD2_v			VSSQ_v	VSSQ_v	VDDQ_v	DQ1_v	VDDQ_v	DQ2_v	
W		CA2_v	CA3_v	VSSCA_v	VDD2_v			VSSQ_v	DQS2_c_v	DM2_v	VDDQ_v	DQ0_v	VSSQ_v	
Y		CA0_v	CA1_v	VSS_v	VSS_v			VDDQ_v	DQS2_t_v	VSSQ_v	DQ23_v	VDDQ_v	DQ22_v	
AA	DNU	VSS_v	VDD1_v	VSS_v	VDD1_v			VSS_v	[illegible]	DQ21_v	VSSQ_v	DQ20_v	DQ19_v	DNU
AB	DNU	DNU	VDD1_v	VDD1_v	VDD2_v			VDD2_v	VDD1_v	DQ18_v	DQ17_v	DQ16_v	DNU	DNU

U2001 Chip Bitmap:

209	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
A	VPA	VSYS_VPA	VSYS_VPA	VSYS_VPROC	GND_VPROC	VPROC	VPROC	VSYS_VCORE	VCORE	VCORE		GND_VMODEM	VSYS_VMODEM	VSYS_VS1	GND_VS1	VS1	VCAMIO	A
B	GND_SMPS	VSYS_SMPS	GND_VPA	VSYS_VPROC	GND_VPROC	VPROC	VPROC	VSYS_VCORE	GND_VCORE	VCORE	VMODEM	GND_VMODEM	VSYS_VMODEM	VSYS_VS1	GND_VS1	VS1	VIO18	B
C		VPROC_FB		VSYS_VPROC	GND_VPROC	VPROC	VPROC	VSYS_VCORE	GND_VCORE	VCORE	VMODEM	GND_VMODEM					VS1_LDO1	C
D	AU_V18N	GND_VPROC_FB	VCORE_FB			D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	GND_VMODEM_FB		VRF18		D
E		FLYN	GND_VCORE_FB	VPA_FB		D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	VMODEM_FB	VS1_FB	VCN18	VRF12	VCAMD	E
F	FLYP	AVSS18_AUD	AU_LOLN	AU_LOLP	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND		VS2_LDO1		VS2_LDO2	F
G		AVDD18_AUD	AU_HPR		AU_HSN	AU_HSP	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND		VSYS_LDO1	VSRAM_PROC	VSRAM_OTHERS		G
H			AU_REFN		AVSS28_AUD	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND	D_GND			VEFUSE	VCAMA	VDRAM	H
J	HP_EINT	AU_HPL		AU_VIN2_P	AU_VIN2_N	D_GND	D_GND	D_GND	D_GND	DVSS18_IO	D_GND			VUSB			VSIM1	J
K	AVDD28_AUD		AU_VIN0_P	AU_VIN0_N	AU_VIN1_P		VSYS_LDO3			DVDD18_IO		D_GND	VCN28		VIO28	VSIM2		K
L			AU_MICBIAS0		AU_VIN1_N		VAUD28			DVDD18_DIG		ISINK1		VFE28	VMC	VLDO28	VEMC	L
M	XTAL1	AVSS22_XO	AU_MICBIAS1	ACCDET	EXT_PMIC_EN1	EXT_PMIC_PG	SPI_MOSI	SPI_CLK	SPI_MISO	CHG_DM	CHG_DP		BATSNS	AUD_SYNC_MOSI		VIBR		M
N	XTAL2	AVSS22_XO_ISO	UVLO_VTH	TCHR_ENB	EXT_PMIC_EN2		SRCLKEN_IN0	SRCLKEN_IN1	VSYSSNS			PCHR_LED	ISENSE	AUD_DAT_MOSI		VMCH	VSYS_LDO2	N
P	AVSS22_XOBUF	AVSS22_XO_ISO	XO_WCN		PMU_TESTMODE		AVSS18_AUXADC		ISOURCE		RTC32K_2V8		VDRV	RTC32K_2V8_0	AUD_DAT_MOSI0	AUD_CLK_MOSI	VCN33	P
R	AVSS22_XOBUF	XO_NFC	XO_SOC	PWRKEY	RESETB		AVDD18_AUXADC	SPI_CSN	TREF	CS_N	CHRLDO	VRTC28	BATON		RTC32K_1V8_1	AUD_CLK_MISO	AUD_DAT_MISO1	R
T	XO_CEL	XO_EXT		VXO22	VAUX18		AUXADC_VIN	WDTRSTB_IN		CS_P	VCDT		GND_VREF	VREF		AUD_SYNC_MISO	AUD_DAT_MISO0	T
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	

A1 CHG_VIN	A2 CHG_VIN	A3 CHG_VIN	A4 D-	A5 D+	A6 FL_LEDCS2	A7 FL_LEDCS1		A9 PD_CC2	A10 BL_VLX
B1 CHG_VMID	B2 CHG_VMID	B3 CHG_VMID	B4 FL_VMID	B5 FL_VMID	B6 FL_VMID		B8 PD_IRQB	B9 PD_VCONN5V	B10 BL_PGND
C1 CHG_VLX	C2 CHG_VLX	C3 CHG_VLX			C6 FL_VINTORCH		C8 CHG_ILIM	C9 PD_CC1	C10 BL_VDDA
D1 CHG_PGND	D2 CHG_PGND	D3 CHG_PGND	D4 CHG_BOOT	D5 VBATS	D6 FL_STROBE	D7 FL_TORCH	D8 CHG_OTG	D9 BL_VOUT	D10 BL_LED1
E1 VSYS	E2 VBUS	E3 CHG_VDDP	E4 OVP_CTRL	E5 AGND	E6 AGND	E7 AGND	E8 CHG_QONB	E9 CHG_ENB	E10 BL_LED2
F1 VSYS	F2 VSYS	F3 CHG_VBATOVPB	F4 NC	F5 AGND	F6 AGND	F7 AGND	F8 CHG_DSEL		F10 BL_LED3
G1 VBAT	G2 VBAT		G4 CHG_STAT	G5 AGND	G6 AGND	G7 AGND	G8 FL_TXMASK	G9 AGND	G10 BL_LED4
H1 VBAT	H2 VBAT		H4 RGB_ISINK4	H5 RGB_ISINK1	H6 DB_ENN	H7 DB_ENP	H8 MRSTB	H9 BL_PWM	H10 BL_EN
J1 LDO_VOUT	J2 LDO_VIN	J3 TS	J4 RGB_ISINK3	J5 RGB_ISINK2	J6 IRQB		J8 SCL	J9 SDA	J10 VDDA
K1 DB_NEGVOUT	K2 DB_NEGCF2	K3 DB_NEGPGND	K4 DB_NEGCF1	K5 DB_POSVOUT	K6 DB_BSTVOUT	K7 DB_POSVLX	K8 DB_POSPGND	K9 VDDM	K10 DB_VDDA

3.1 Switching failure

In the process of maintenance without starting up, we should follow the principle of software before hardware, pay attention to observe whether the motherboard components are damaged, breakdown, liquid inflow, etc. In specific measurement, we should measure according to the starting time sequence.

3.1.1 Fix the screen without starting up

Analysis ideas:

1. Software upgrade, troubleshooting software.

2. If the software upgrade reports an error, measure whether the working conditions of U1001 and U4102 are normal.
3. If the software is still constant after upgrading, measure whether the startup timing and CPU working conditions are normal.

Maintenance Case 1

Fault phenomenon: fixed screen

Faulty Component: Software Upgrade

Maintenance method: After starting up, it will be set in the interface of "The system has been destroyed", and the mobile phone system will be damaged, and the fault will be repaired after software upgrade.

Maintenance Case 2

Fault Phenomenon: White Rice Fixed Screen Fault Component: Software Upgrade

Maintenance method: start up and fix the screen with white rice, and repair the fault after software upgrade.

Maintenance Case 3

Fault phenomenon: White rice fixed screen fault cause: U4102

Maintenance method: start-up fixed white rice, invalid software upgrade, no problem in measurement, and repair the fault after replacing U4102.

Maintenance Case 4

Failure phenomenon: Android fixed screen failure cause: U4102

Maintenance method: Start Android, the software upgrade is invalid, and the fault is repaired after replacing U4102.

3.1.2 Constant current without startup

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. If the software upgrade reports an error, measure whether the working conditions of U1001 and U4102 are normal.
3. If the software is still constant after upgrading, measure whether the startup timing and CPU working conditions are normal.

Maintenance Case 1

Fault phenomenon: 60mA-70mA constant current fault cause: software upgrade

Maintenance mode: 60mA-70mA constant current does not start up, and the fault is repaired after software upgrade.

Maintenance Case 2

Fault phenomenon: 70mA constant current fault cause: U4102

Maintenance mode: 70mA constant current, software upgrade error "STATUS-DA-HASH-MISMATCH", fault repair after replacing U4102.

Maintenance Case 3

Fault phenomenon: 80mA constant current fault cause: U4102

Maintenance mode: 80mA constant current, software upgrade reports error "disaster", and fault repair after replacing U4102.

3.1.3 No startup current is not maintained

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether the startup timing signal is normal.
3. Measure whether the power supply of U1001 is normal.
4. Measure whether the USB signal circuit is normal (USB_ID; USB_DM; USB_DP, note: if these measured signals are abnormal, CPU fault can be judged from the side).

Maintenance Case 1

Fault phenomenon: 50mA ~ 100mA jump is not maintained Fault cause: U4102

Maintenance mode: the jump of startup current from 50mA to 100mA is not maintained, the software upgrade reports an error of "STATUS-DA-HASH-MISMATCH", and the fault is repaired after replacing U4102.

3.1.4 No startup, no current

Analysis ideas:

1. Check whether the appearance of J0002 is damaged, and if the interface is normal, measure whether the ground value of J0002 is normal (BAT_CON_ID, BAT_THERM, VBATT).

2. Check whether J3 is damaged and measure whether the voltage of PWRKEY is normal.
3. Measure whether the startup timing signal is normal.

3.1.5 Leakage without starting up

Maintenance ideas:

1. First of all, visually check whether there are components rupture, breakdown, liquid corrosion and discoloration in the appearance of the motherboard.
2. Measure whether VBAT and VSYS are short-circuited. If both VBAT and VSYS are short-circuited, find out the VBAT short-circuited element first, and then find the VSYS short-circuited element.
3. Measure whether there is short circuit in other power supply lines, and find out the faulty components according to the short circuit signal.
4. Power on to find heating elements.

Maintenance Case 1

Fault Phenomenon:
Leakage 960mA Fault
Component: U3310

Maintenance method: 960mA electric leakage can be turned on, visual inspection of U3310 breakdown, replacement of U3310 after the fault repair.

Maintenance Case 2

Fault Phenomenon:
Electric Leakage 1.32 A
Replacement
Component: U3310

Maintenance method: leakage of 1.32 A, measurement of Vbat voltage short circuit, visual inspection of U3310 breakdown, replacement of U3310 fault repair.

Maintenance Case 3

Fault Phenomenon:
Leakage 510mA Fault
Component: U3310

Maintenance method: leakage of 510mA, measurement of Vbat voltage short circuit, replacement of U3310 fault repair.

Maintenance Case 4

Fault Phenomenon:
Leakage 3.05 A Fault
Component: U2001

Repair method: Electric leakage 3.05 A, measuring VSYS voltage short circuit, U2001 heat serious after replacement fault repair.

Maintenance Case 5

Fault Phenomenon:
Leakage 1.78 A Fault
Component: U2501

Maintenance methods: leakage of 1.78 A, measurement of Vbat voltage short circuit, U2501 heat serious replacement after the fault repair.

3.2 Restart failure

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether I2C to ground value and voltage are normal.
3. Measure whether the clock is normal when U1001 is powered.
4. Measure whether the motherboard has abnormal power supply caused by short circuit.
5. Replace U1001.

Maintenance Case 1

Fault phenomenon:
White rice restarts
faulty components:
software upgrade

Maintenance method: Power-on has been restarted in white rice, and the fault is repaired after software upgrade.

Maintenance Case 2

Failure Phenomenon:
Automatic Restart
Failure Mode: U1001

Maintenance method: standby automatic restart, touch screen failure, invalid software upgrade, no problem in measurement, and repair the fault after replacing U1001.

3.3 Crash failure

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether BAT_ID and BAT_THERM are normal.
3. Measure whether the startup timing is normal.
4. Measure whether the power supply of U1001 is normal.
5. Measure whether there is abnormal power supply caused by short circuit on the motherboard.
6. Replace U1001.

Maintenance Case 1

Failure Phenomenon:
Standby Crash Failure
Cause: U1001

Maintenance method: standby crash, invalid software upgrade, fault repair after replacing U1001.

3.4 Signal fault

Analysis ideas:

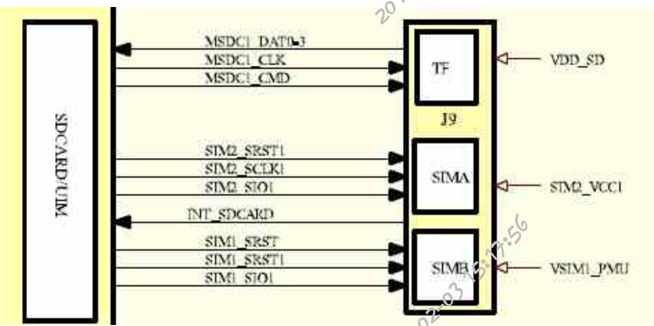
1. Insert SIM card to ensure normal identification, and eliminate faults that do not recognize SIM card.
2. Software upgrade, troubleshooting software.
3. RF calibration, check the specific test items through the test report. However, measure the RF path according to the corresponding system and principle block diagram to find the fault point.
4. Measure whether the power supply of RF circuit is normal.
5. If the RF calibration is normal and there is still no signal, check the SIM card circuit.

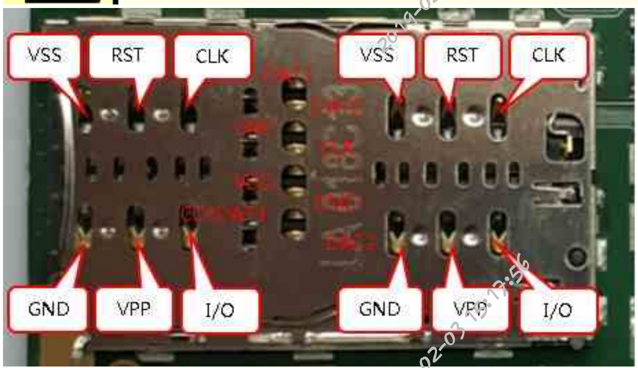
Maintenance Case 1
Fault phenomenon:
weak signal fault
cause: software
upgrade

Maintenance method: weak signal, calibration OK after software upgrade, fault repair.

3.5 SIM card failure

Principle block diagram





SIM/SD测量表		
Symbol	测量值	测量点
VSS	800	SIM1
RST	800	
CLK	800	
VPP	800	
I/O	550	
VSS	800	SIM2
RST	800	
CLK	800	
VPP	800	
I/O	680	
DAT1	800	SD
DAT0	800	
CLK	800	
VDD	1	
CMD	800	
CD/DAT3	800	
DAT2	800	

1. First, check whether the baseband version of the mobile phone is normal. If the baseband information is normal, it is the SIM card related function failure.
2. Check whether the SIM card pin is deformed, oxidized or broken, and carefully observe whether there is virtual welding phenomenon in the solder joint of SIM card holder.
3. Measure whether the SIM card aims at the ground value normally.
4. Start up and test whether the power supply, clock and reset of SIM card are normal, and whether the data voltage jump is normal
5. If the above signal is normal, replace U1001.

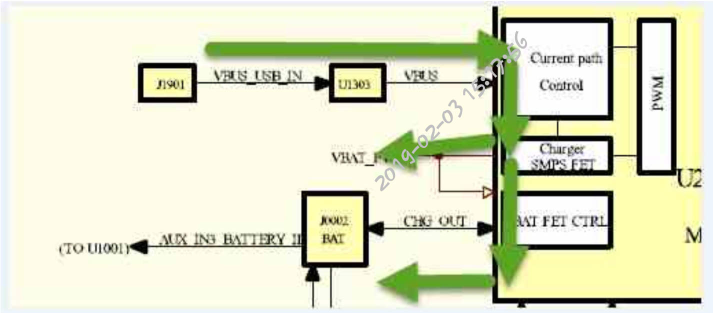
Maintenance Case 1
Fault Phenomenon:
Ignorance of SIM Card
Fault Component: J9

Maintenance analysis: I don't know SIM card, J9 is damaged by visual inspection, and the fault is repaired after replacing J9.

Maintenance Case 2
Fault Phenomenon: Not aware of SIM card Fault
Component: U3101
Maintenance method: SIM card is not recognized, baseband version shows "unknown", software upgrade and welding U1001 are invalid, card recognition is normal after replacing U3101, signal is normal, calibration passes fault repair.

3.6 Charging function failure

Charging schematic diagram:



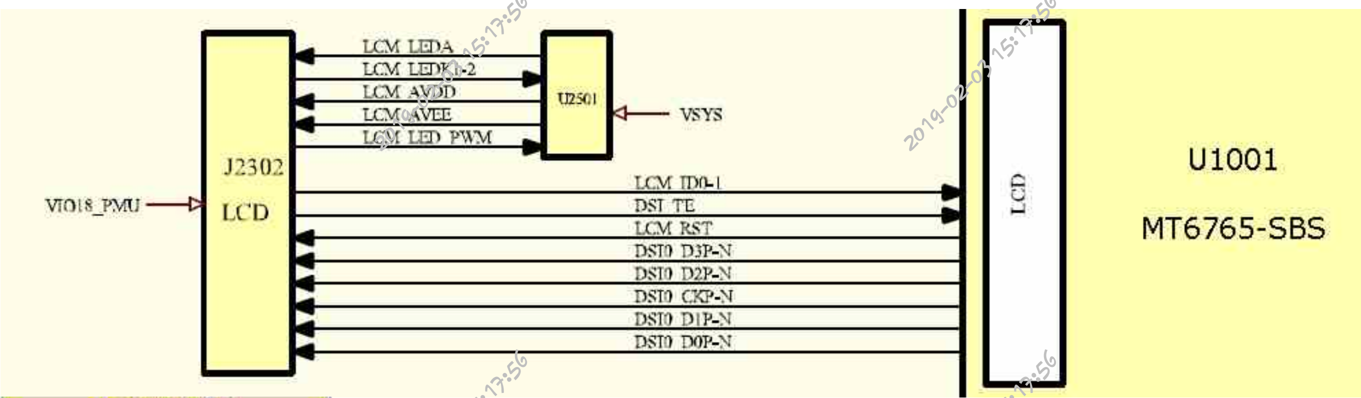
1. Check whether the appearance of J0002 and J1901 is normal, and measure whether the ground values of VBUS, USB_PMI_DP and USB_PMI_DM signals are normal with multimeter diode gear.

Maintenance Case 1
Fault phenomenon:
Abnormal electric quantity display Fault component:
U2501
Maintenance method: The electric quantity display is abnormal, and the fault is repaired after replacing U2501.

Maintenance Case 2
Fault Phenomenon:
Slow Charging
Fault Component:
U2501
Maintenance method: It can be charged, the charging current is low, and the fault is repaired after replacing U2501.

3.7 Display fault

Principle block diagram



显示电压测量表		
Symbol	测量值	测量点
LCM_LEDA	4V-29V	R2706
LCM_LEDA_1	0.6V	C22
LCM_LEDA_2	0.6V	C39
LCM_AVDD	5V	R43
LCM_AVEE	-5V	R44
LCM_RST	1.8V	C32
VIO18_PMU	1.8V	C34

Maintenance ideas:

1. Visually check whether J2302 and peripheral components are damaged or falsely welded.
2. Brush the machine to eliminate software faults.
3. Measure whether the ground value of each pin of J2302 is normal with multimeter diode gear.
4. If the ground value is normal, measure whether the power supply and control signals in the "display meter" are normal.
5. Replace U1001.

Maintenance Case 1
Fault Phenomenon:
Black Screen Fault
Component:
U1001

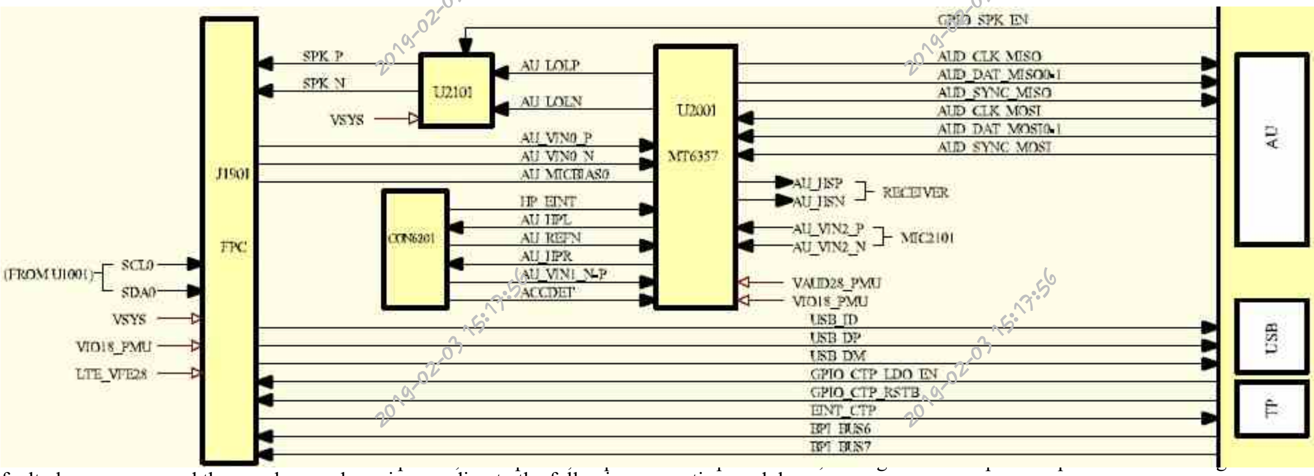
Maintenance method: The mobile phone is turned on with a black screen directly, and U1001 is replaced for fault repair.

Maintenance Case 2
Fault Phenomenon:
No Display Fault
Component: U2501

Maintenance method: No display, measuring LCM_AVDD, LCM_AVEE without positive or negative 5V voltage, and repairing the fault after replacing U2501.

3.8 Audio fault

Principle block diagram



fault phenomenon, and then analyze and repair according to the following respective modules.

3.8.1 Loudspeaker failure

Speaker is connected to the motherboard through FPC. Its principle is to go to CODEC through CPU first, and then amplify and output to interface J1901 through U2101 audio power amplifier and then to speaker. Maintenance ideas:

1 Visually check whether J1901 looks good.

2. Software upgrades troubleshoot software.
3. Measure whether SPK + and SPK-to-ground values are normal with multimeter diode gear.
4. Measure whether U2101 voltage is normal.
5. If the above signals are normal, consider whether the bus from CODEC to CPU is normal and whether the speaker path is open.

Maintenance Case 1
Fault Phenomenon:
Loudspeaker Noise Fault
Component: U2101

Maintenance method: loudspeaker noise, invalid software upgrade, fault repair after replacing U2101.

3.8.2 MIC fault

Redmi 6 & Redmi 6A motherboard contains two MIC circuits, main/auxiliary MIC, the main MIC is lead type and welded on the auxiliary board. Maintenance ideas:

1. Visually check whether J1901 looks good.
2. Measure AU_VIN0_P; AU_VIN0_N; AU_MICBIAS0 is normal for ground value.
3. Measure whether the AU_MICBIAS0 voltage is normal.
4. If the above signals are normal, consider whether the bus between CODEC and CPU is normal.

3.8.3 Receiver failure

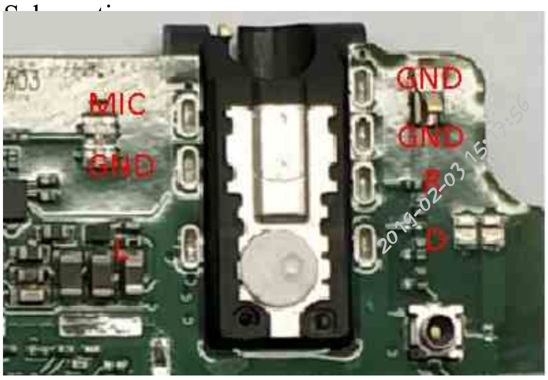
Maintenance ideas:

1. Visual inspection eliminates appearance problems.
2. Software upgrade, troubleshooting software.
3. Measure whether the installation position of the motherboard earpiece is normal to the ground value.
4. According to the audio signal trend of REC measured in the circuit diagram, reverse analysis is carried out.

3.8.4 Headphone failure

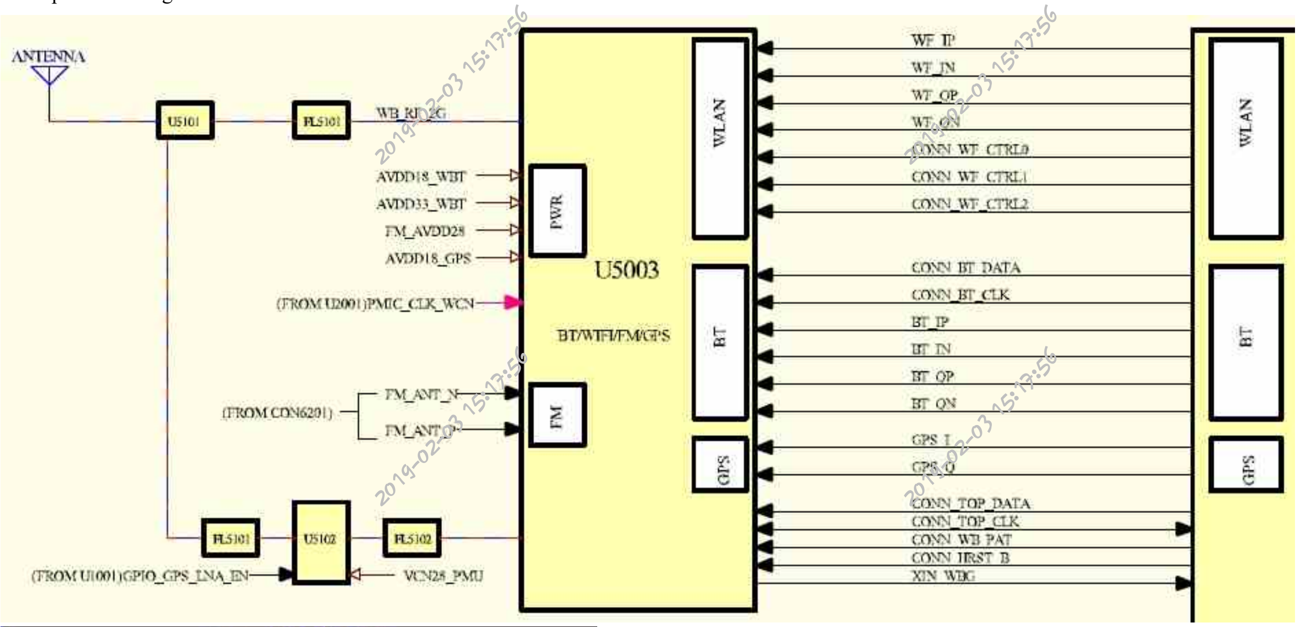
Maintenance ideas:

1. Check whether the earphone interface contacts are deformed and whether there are foreign bodies in the earphone interface.
2. Software upgrade, troubleshooting software.
3. Measure whether the earphone interface to the ground value is normal, do not insert the earphone when starting up, and measure whether the DET detection signal has 1.8 V voltage.
4. Repair the headphone channel according to the principle block diagram, and replace U1001 normally.
5. Replace U1001.



3.9 WIFI/BT/FM/GPS Fault

Principle block diagram



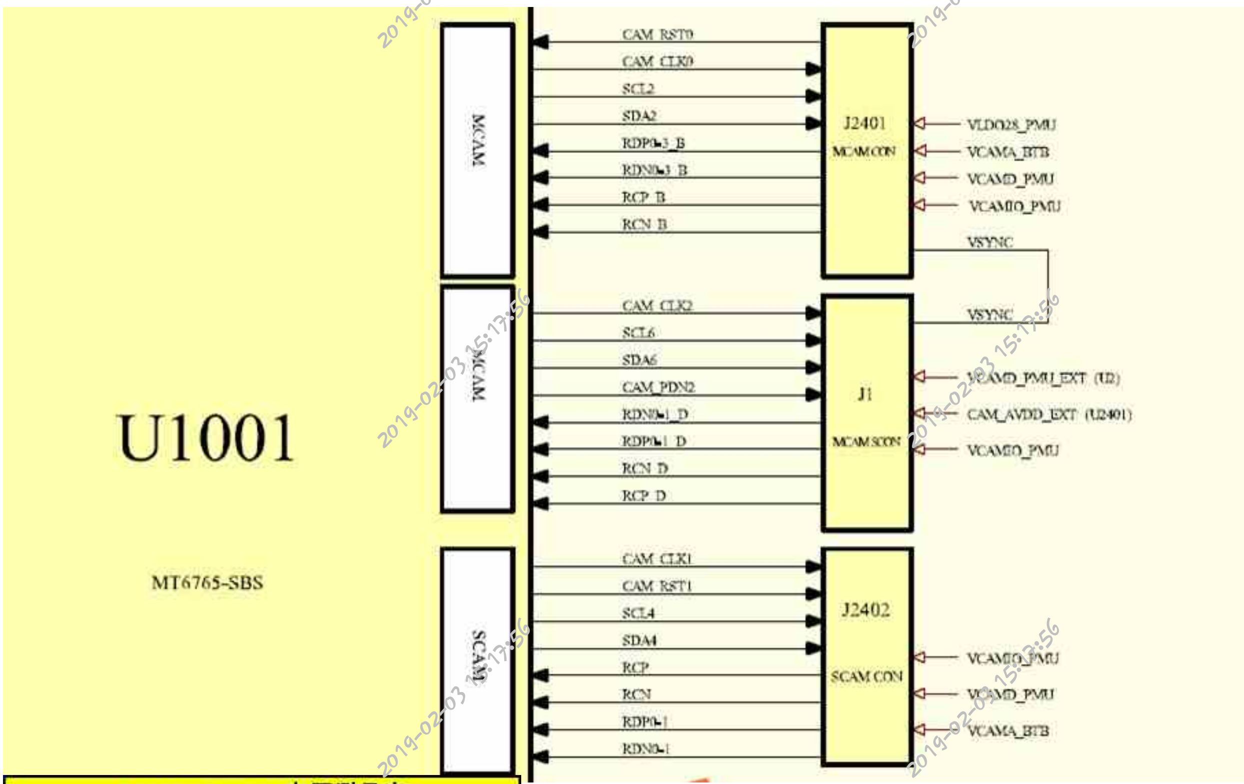
U5003电压测量表		
Symbol	测量值	测量点
FM_AVDD28	2.8V	C5019
AVDD18_GPS	1.8V	R5008
AVDD18_WBT	1.8V	R5012
AVDD33_WBT	3.3V	R5005

Maintenance ideas:

1. Software upgrade and troubleshooting.
3. Measure whether the power supply, clock and enable signals of U5003 are normal.
4. Take off U5003 and measure whether the bus between U1001 is normal. If it is normal, replace U5003.
5. Replace U1001.

3.10 Camera failure

Principle block diagram



Front Camera电压测量表		
Symbol	测量值	测量点
VCAMA_BTB	2.8V	C108
VCAMIO_PMU	1.8V	C2403
VCAMD_PMU	1.2V	C2412

Rear Main Camera A signal measurement:

Main Camera A电压测量表		
Symbol	测量值	测量点
VLDO28_PMU	2.8V	C2409
VCAMD_PMU	1.2V	C2411
VCAMIO_PMU	1.8V	C2408
VCAMA_BTB	2.8V	C2410

Rear Main Camera B signal measurement:

Main Camera B电压测量表		
Symbol	测量值	测量点
CAM_AVDD_EXT	2.8V	C26
VCAMD_PMU_EXT	1.2V	C27
VCAMIO_PMU	1.8V	C24
CAM_PDN2	1.8V	TP9

Maintenance ideas:

1. Software upgrade, troubleshooting software.
2. Detect J2402, J2401, J1 and surrounding components for loss and damage.
3. Go to CIT to test the front camera and the rear camera to distinguish faults.
4. Measure whether the ground values of J2402, J2401 and J1 are normal.
5. Measure whether the camera power supply, clock and reset signal output are normal.
6. Measure whether the I2C and MIPI bus output by U1001 are normal.

Maintenance Case 1
Fault Phenomenon:
Faulty Component:
U1001 Can't be opened
after shooting

Maintenance method: The measurement signal PDN2 is short-circuited, and the fault is repaired after replacing U1001.

Maintenance Case 2
Fault phenomenon: Black
screen fault component of
rear camera: EM2407

Maintenance method: Take a black screen after taking photos, measure the abnormal resistance at one end of RDP3, and repair the fault after replacing EM2407.

Maintenance Case 3
Fault Phenomenon: Front
Photography Black Screen
Fault Component: U1001

Maintenance method: front photo black screen, measurement of RDN0 resistance infinity, replacement of U1001 after the fault repair.

Maintenance Case 4
Fault Phenomenon: Front
Photography Black Screen
Fault Component: J2402

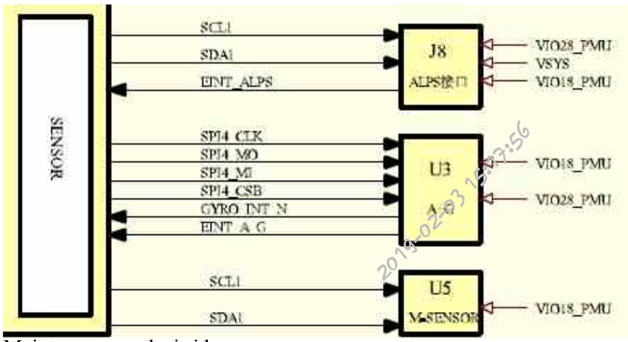
Maintenance method: Take a black screen in front of the camera, measure the ground of J2402, short circuit of the eighth pin to the ground, and repair the fault after replacing J2402.

Maintenance Case 5
Fault Phenomenon: Fault
Component: U1001

Maintenance method: The front can not be opened, the resistance of CAM_CLK1 is measured to be infinite, and the fault is repaired after replacing U1001.

3.11 Inductor failure

Principle block diagram



Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether the working conditions of the corresponding sensors are normal.
3. Replace the corresponding sensor.
4. Replace U1001.

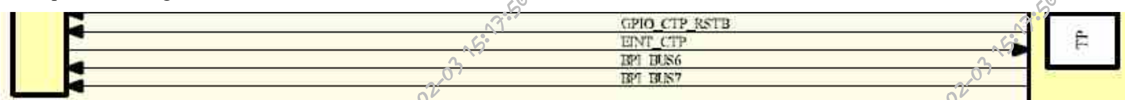
Maintenance Case 1
Fault phenomenon: Gravity
induction failure Fault
component: U5

Maintenance method: gravity induction failure, invalid software upgrade, fault repair after replacing U5.

Maintenance
Case 2
Failure
Phenomenon: Failure
Component:
Maintenance
Method:

3.12 Touch screen failure

Principle block diagram:



Maintenance analysis ideas:

1. Check whether J1901 and surrounding components are damaged (touch screen connection sub-board is connected to motherboard through FPC).
2. Software upgrade, troubleshooting software.
3. Measure whether the working conditions are normal.
4. Replace U1001.

Maintenance Case 1
Fault phenomenon:
touch screen failure
Fault component:
U1001

Maintenance method: touch screen failure, measure SDA0 short circuit to ground, replace U1001 after the fault repair.

Maintenance Case 2

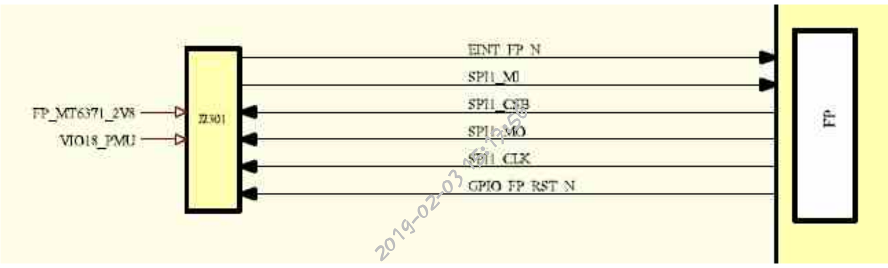
Failure phenomenon: touch screen failure

Faulty component: U1001

Maintenance method: touch screen failure, measure SCL0 short circuit to ground, and repair the fault after replacing U1001.

3.13 Fingerprint identification failure

Principle block diagram



指纹电压测量表		
Symbol	测量值	测量点
VIO18_PMU	1.8V	C2311
FP_MT6371_2V8	2.8V	TP13
EINT_FP_N	1.8V	R7

Maintenance analysis ideas:

1. Visually check whether J2301 and its surrounding components are damaged. If there is damage, please replace it.
2. Brush the machine to eliminate software faults.
3. Measure whether the above voltage and other signals are normal.
4. If all the above signals are normal, replace U1001.